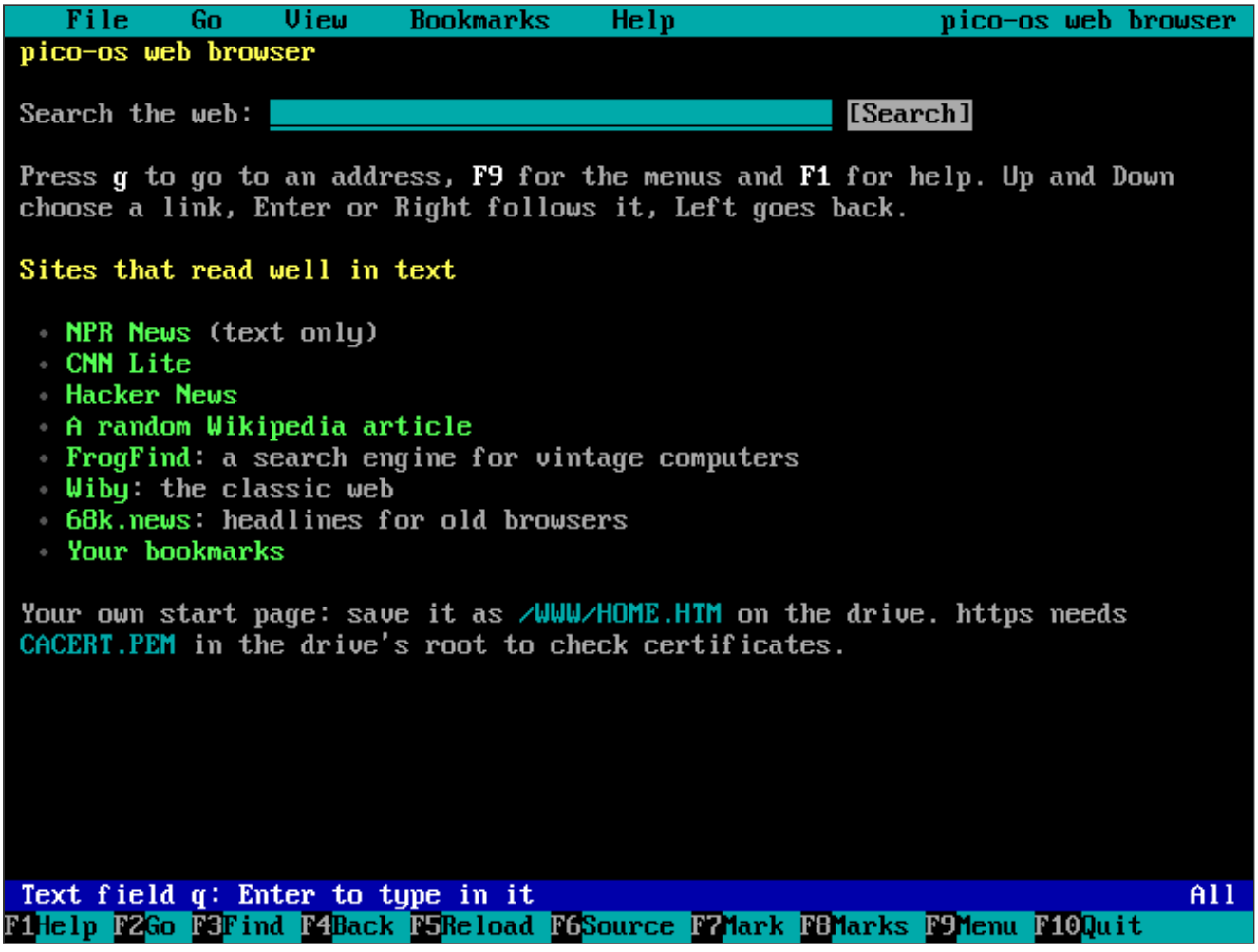
Pico 2 mini OS Web Browser

Pico 2 mini OS · October 30, 2026

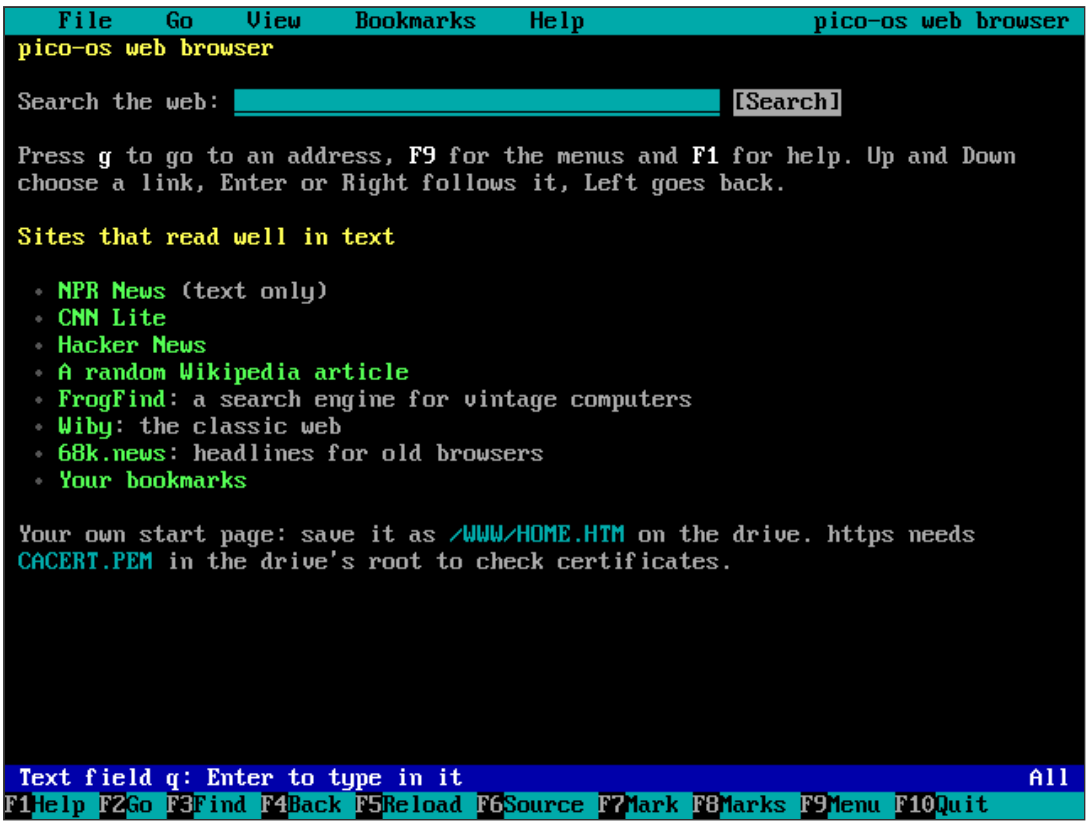


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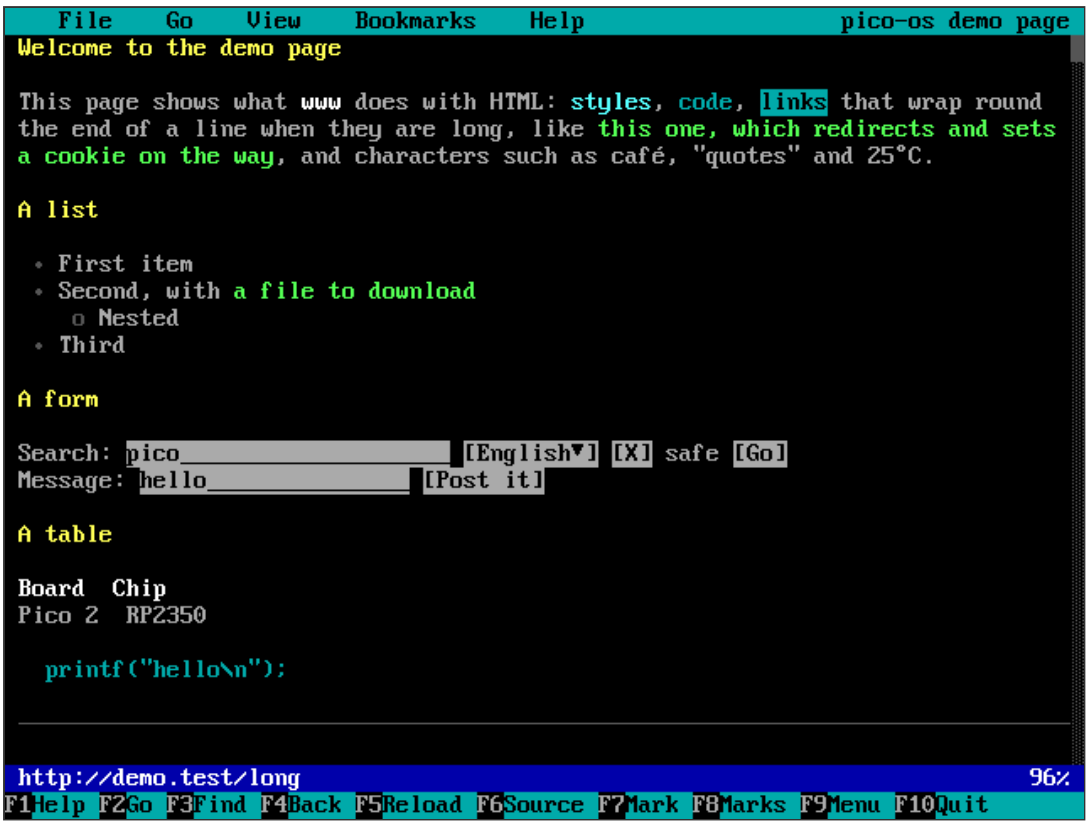
`www` (or `lynx`) is a text web browser in the manner of lynx: it loads pages over http and https (the same network and TLS code as `wget`) and shows them as text on the 80x30 screen, with their links and simple forms, and with its pictures drawn between the text; there are no scripts or styles; most news, reference and search pages read well. It needs WiFi (`wifi connect`), and for https the certificate file `CACERT.PEM` in the drive's root (as for `wget`).

Starting



- `www` opens the start page: `/WWW/HOME.HTM` on the drive if there is one, else a built-in page with a search box and some sites that work well in text.
- `www text.npr.org` (or `www https://text.npr.org/`) opens that address. Without `http://` or `https://`, https is used.
- `www /DOCS/PAGE.HTM` opens a page on the drive (also `file:///...`).
- `www raspberry pi pico` searches the web for those words (DuckDuckGo's HTML page).

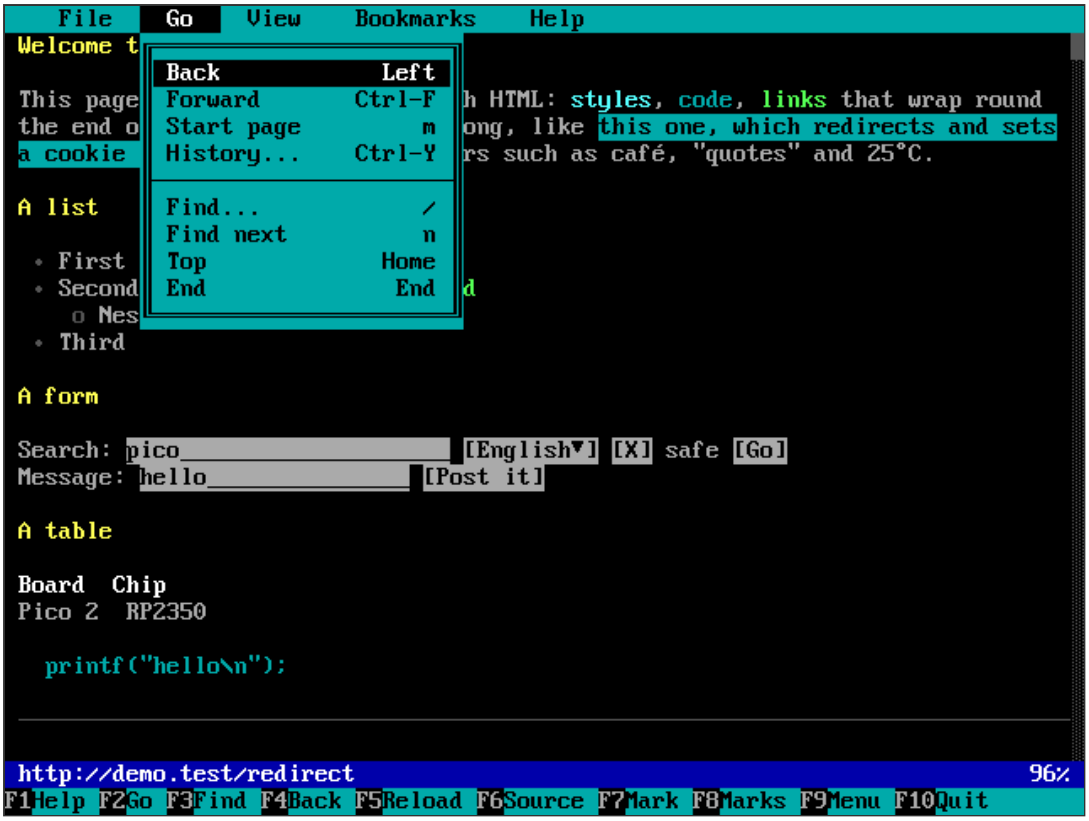
The screen



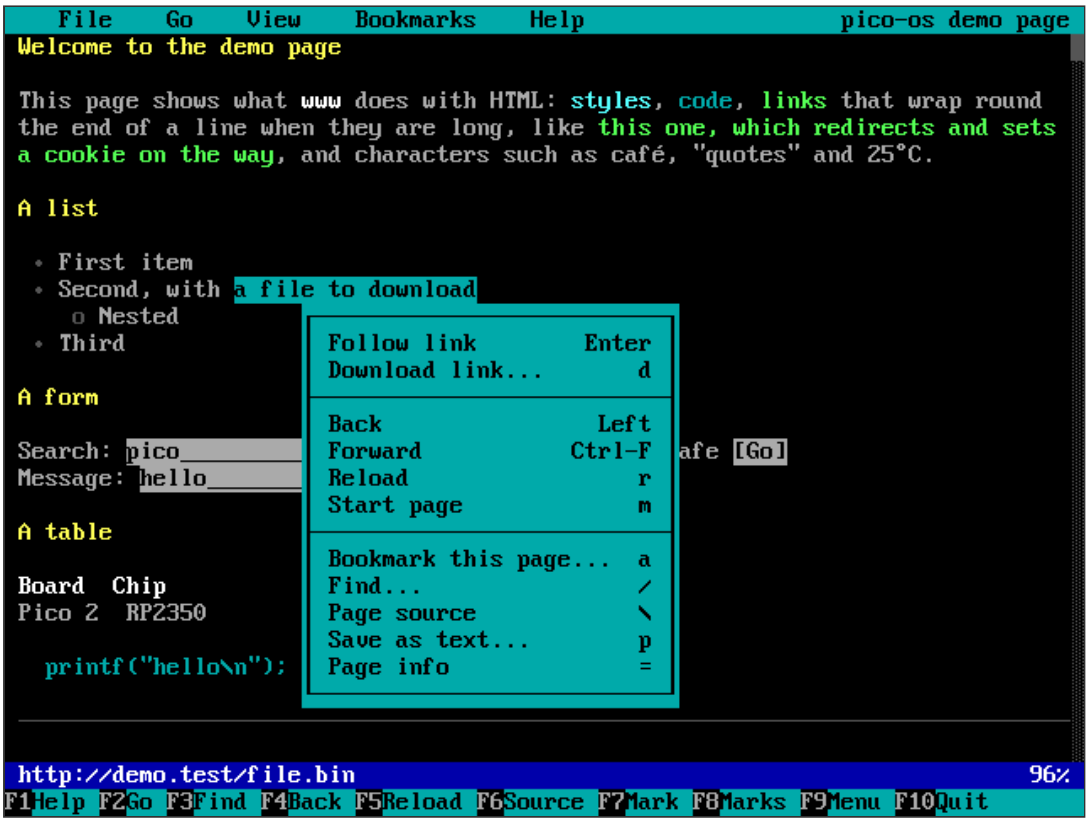
- **The menu bar** at the top: File, Go, View, Bookmarks and Help, as in mc and the tracker. F9 (or Alt and a title's first letter, or a click) opens it. The page's title is on the right.
- **The page**, 78 characters wide. Headings are yellow, bold text white, italics light cyan, code cyan, a picture's alt text magenta in `[brackets]`. **Links are green**; the link chosen is black on cyan. Form fields are grey, with `____` for the room in a text field, `[X]` for a checkbox, `(*)` for a radio button and `[choice▼]` for a list.
- **The scroll bar** on the right (when the page is longer than the screen): click it to jump there.
- **The status line** says where the chosen link goes (or what the form field is), and how far down the page is shown. While a page loads it shows the progress: looking up the name, connecting, the TLS handshake, the kilobytes so far. Esc stops loading and shows what arrived.
- **The key bar** at the bottom: click a key to press it.

Keys

Key	
Up, Down, Tab	The previous or next link. When the next one isn't on the screen, the page scrolls half a screen.
Enter, Right	Follow the link, or work the form field (below).
Left, Backspace, F4	Back. Ctrl-F goes forward again.
PgDn, Space / PgUp, b	A screen down / up.
Home / End	The top / the end of the page.
g	Go to an address (or type words to search for them).
G, F2	Edit this page's address and go there.
o	Open a file on the drive.
/, F3	Find text on the page (any case); **n** finds it again (from the top after the end).
\, F6	The page's source (the HTML as it came), and back.
r, F5	Load the page again.
m	The start page.
a, F7 / v, F8	Bookmark this page / show the bookmarks.
d	Download the chosen link's file to the drive.
s / p	Save the page's source / save it as text (as shown).
=	About the page and the chosen link.
i	Show the chosen picture, or the chosen link's picture, on the whole screen.
*	Pictures drawn in pages, or [alt text] only.
e	Every menu shown open, or each closed ([+]) until opened.
Ctrl-Y	The history: pick a page to go back (or forward) to.
h, F1	Help (the keys).
q, F10	Quit.



The mouse: click a link to follow it; the wheel scrolls; click the menu bar, the key bar or the scroll bar.



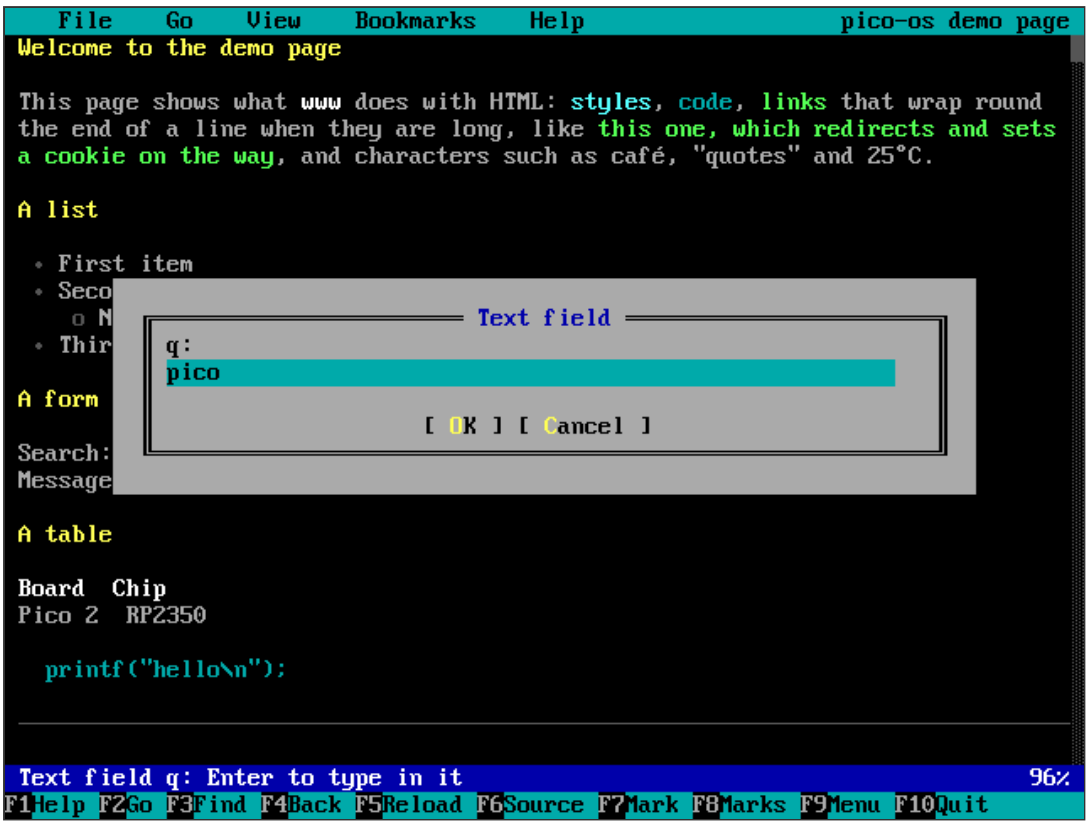
The right button opens a menu where you click: Back, Forward, Reload, Start page, Bookmark this page, Find, Page source, Save as text and Page info, and over a link also Follow link and Download link (the link is chosen), and View image when it is a picture or has

one. Click an item, or hold the right button, move to an item and let go; Up, Down and Enter work too; Esc or a click outside closes it.

Menus

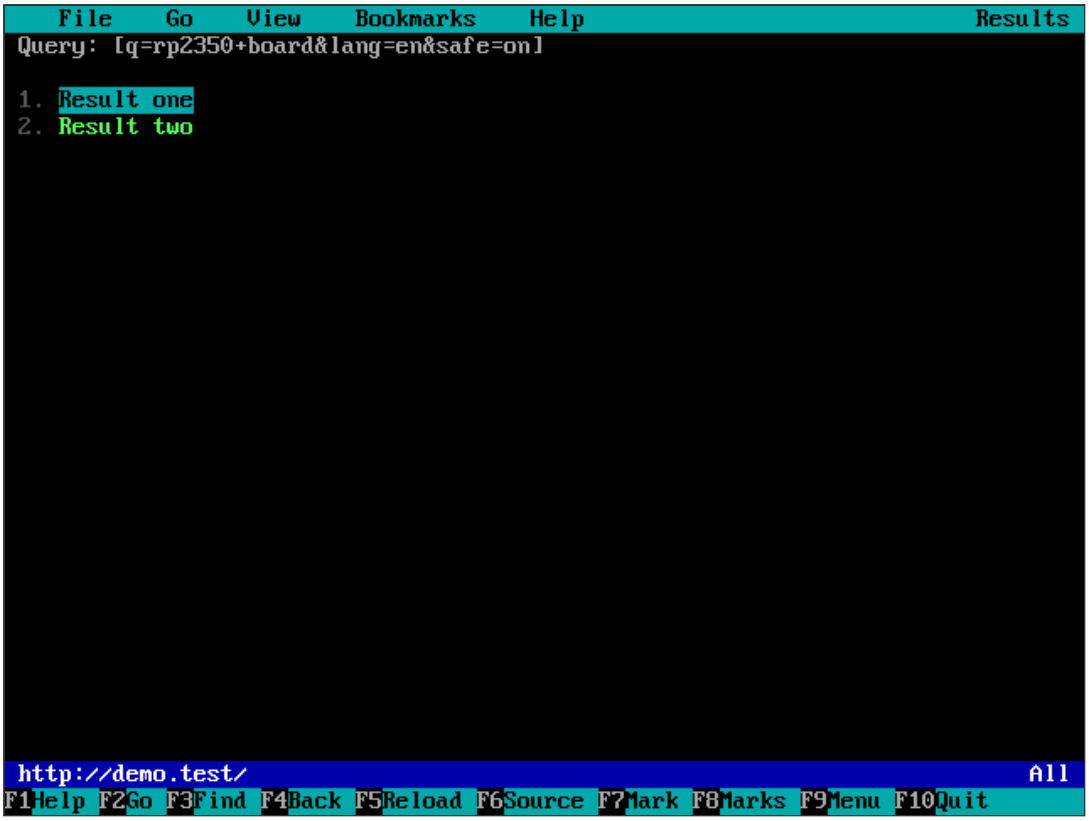
Pages' menus (`<nav>` blocks, and drop-down menus such as Wikipedia's and Fox News' Games and Deals) and `<details>` sections start closed, as one line each: **[+] Main menu**, **[+] 65 languages**, **[+] More info**. They are links: Enter (or a click) opens one, **[-]**, and again closes it; the page keeps its place. So a page starts with what it is about, not its menus. **e** (or View > Open all menus) shows every menu open, as before, and **e** again closes them. Buttons that only work with scripts ("Expand/Collapse Menu") aren't shown. A `<details open>` starts open. Each page's menus start closed again when it is loaded.

Forms



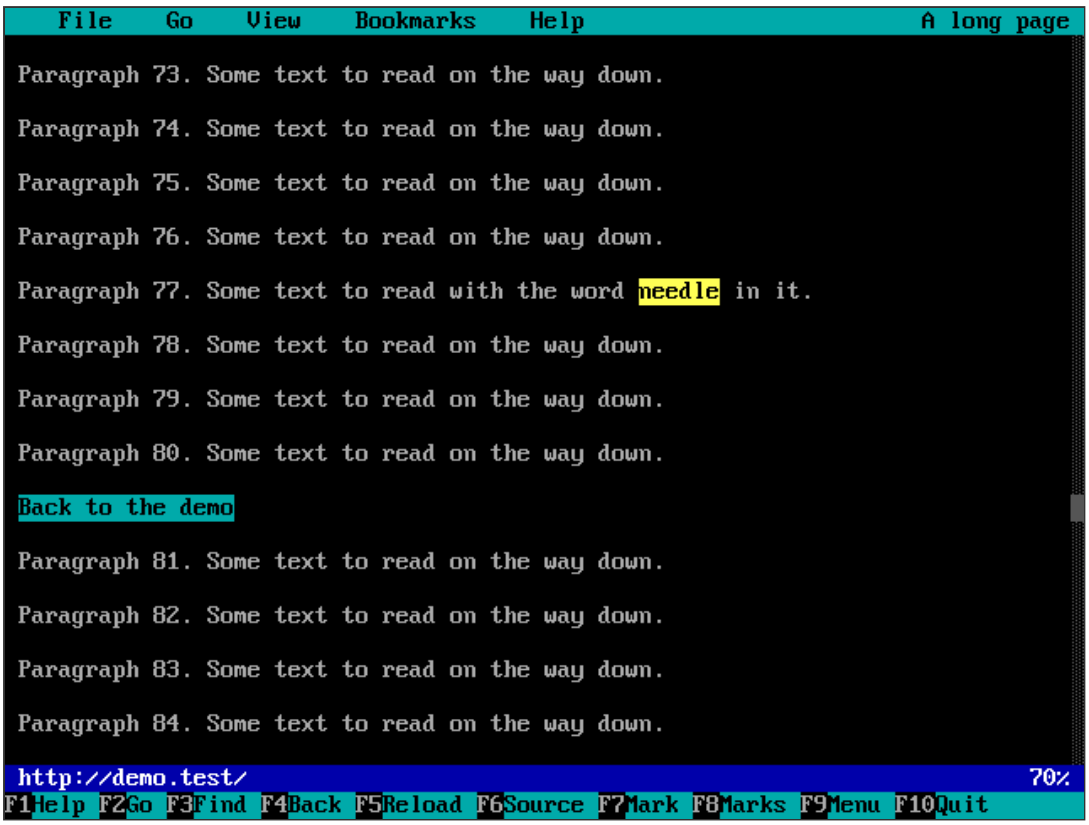
Search boxes and simple forms work. Choose a field with Up and Down, then Enter:

- **A text field** (or a password, or a text area) opens a box to type in (Ctrl-U clears it). In a form with only one text field, such as a search box, Enter in the box sends the form at once.
- **A checkbox** is ticked or cleared; **a radio button** is chosen (the others of its group are cleared).
- **A list** opens the choices to pick from.
- **A button** (`[Search]`, `[Go]`) sends the form, by GET or POST as the page asks.



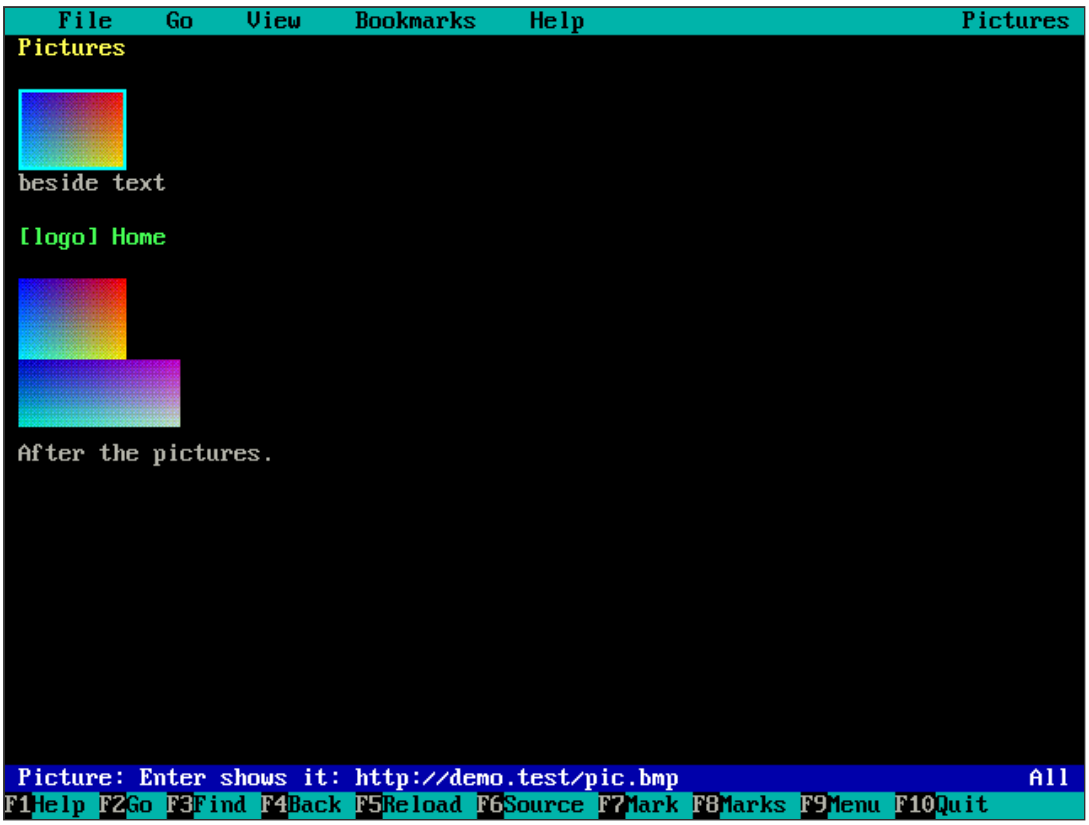
Cookies a site sets are kept (in PSRAM) until the browser closes, so logins and sites that need a cookie work while it is open.

Finding text



`/` asks for the text; the page scrolls to it and it is shown in yellow (and a link on that line is chosen). `n` finds the next one.

Pictures



Pages are text at first, each picture its `[alt text]`. With * (or View > Pictures in pages) pictures (PNG, JPEG and BMP) are drawn in the page, between the text, as they come onto the screen: each loads while no key is pressed (the status line says which; Esc stops it), scaled to fit the page's width and at most 20 lines high, dithered to the screen's 256 colours, never made bigger. A picture whose size the page doesn't give gets a box of a guessed size first; once it has loaded, the page is laid out again around its real size (the place on the page is kept). The mouse pointer is an arrow while a page has pictures.

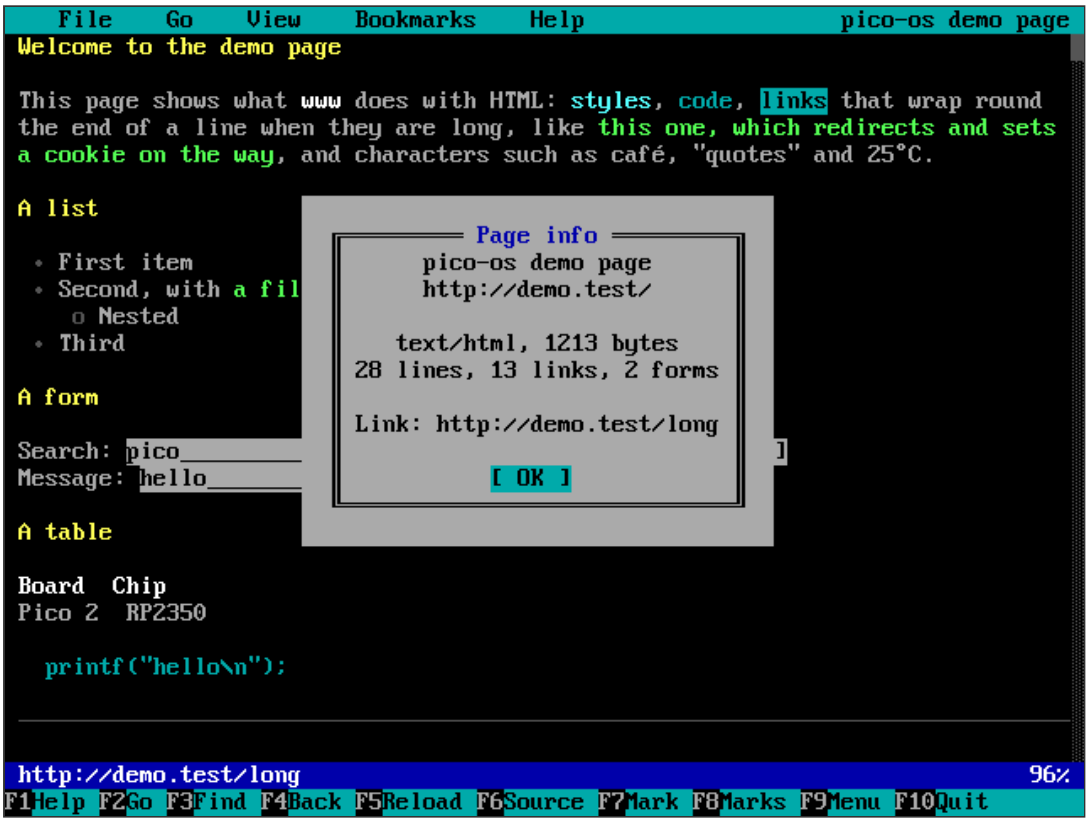
A picture's box shows its `[alt text]` until it is drawn, and stays text if it can't be: GIF, WebP and SVG pictures, tiny icons (40 pixels or less), ones that fail to load, and very big ones. * again goes back to text only; the setting lasts while the browser is open.

A picture in a link follows the link when clicked or chosen with Enter (a cyan frame shows it is chosen); one not in a link is a link of its own (magenta when it is text): Enter (or `i`) shows it on the whole screen, as the `view` command shows a file. A link with a picture in it says `(i: its picture)` on the status line: `i` shows the picture, Enter still follows the link. The right button's menu has **View image** for both. Any key or click goes back to the page.

Pictures drawn are kept in PSRAM (up to 3 MB for the page shown; the ones furthest off the screen go first and load again when they come back), and their sizes are remembered while the browser is open.

Bookmarks, saving and downloading

- **a** adds the page to `/WWW/BOOKMARK.HTM` (an HTML page of links: edit it with `edit` if you like); **v** opens it.
- **d** (or the right button's **Download link**) downloads the chosen link to the drive, page or not. Following a link to something that isn't a page (a ZIP, a ROM, a program, a picture) downloads it too. Either way it asks for the name first, suggesting the one the server gives (or the last part of the address), and asks before replacing a file. The file is written to the drive as it comes, any size, with the progress and speed on the status line; Esc stops it and deletes the part saved.
- **s** saves the page's HTML, **p** the page as text (CRLF lines, as on the screen).
- `/WWW/HOME.HTM` on the drive replaces the start page: any HTML page, for example your bookmarks copied there.



What it does and doesn't do

- HTML is read in one pass: paragraphs, headings, lists (numbered and nested), definition lists, quotes, `<pre>` text, rules, tables (each row on a line, the cells side by side), links, anchors (`#name`), frames and meta refreshes (as links), forms, entities and UTF-8 or Latin-1 text. Characters the screen's font (code page 437) lacks are shown as the nearest one (accents dropped, curly quotes straight) or `?`.
- Not done: pictures beside the text (they are drawn between lines), GIF and WebP and SVG pictures, JavaScript, CSS, layout tables side by side, text areas of more than one line, file uploads, compressed replies (the browser doesn't ask for them), HTTP/2.
- Pages up to 4 MB are loaded; bigger ones are cut there.
- https: the certificate is checked against `CACERT.PEM` (see the *Network Tools* manual). If it can't be checked, the browser asks whether to load the page anyway (Yes, No, or Always for this session); View > Check certificates turns checking off and on.
- It identifies itself as lynx, so sites send their simple pages.

Memory

Everything the browser keeps is in PSRAM, and only while it is open: the pages it has shown (up to 3 MB of them, so Back doesn't load them again; older ones are loaded again when needed), each page laid out, the history (40 pages), the cookies. While a page loads, the network uses the same SRAM as `wget` (a 16 KB receive buffer, and about 40 KB more for https).

Testing on a PC

`tools/wwwtest/uitest.sh /absolute/out/folder` builds the browser for a PC with its real screen, menus, dialogs and fetcher against a pretend web server, drives it with keys and the mouse (pages, links, the right button's menu, pictures, a redirect that sets a cookie, the forms by GET and POST, find, the source, a download, bookmarks, the history, help), checks what the screen shows, and writes the screenshots in this manual (`www_*.png`). It also lays out thousands of random and cut-off pages and checks URLs are resolved as browsers do, all under AddressSanitizer.